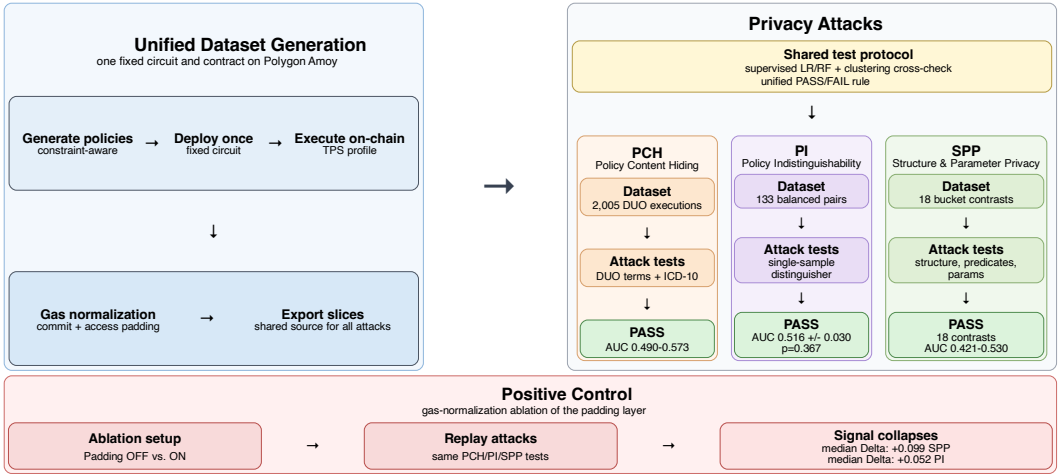

Unified Dataset Generation
one fixed circuit and contract on Polygon Amoy
Generate policies
constraint-aware
Deploy once
fixed circuit
Execute on-chain
TPS profile
Gas normalization
commit + access padding
Export slices
shared source for all attacks
Privacy Attacks
Shared test protocol
supervised LR/RF + clustering cross-check
unified PASS/FAIL rule
PCH
Policy Content Hiding
Dataset
2,005 DUO executions
Attack tests
DUO terms + ICD-10
PASS
AUC 0.490-0.573
PI
Policy Indistinguishability
Dataset
133 balanced pairs
Attack tests
single-sample distinguisher
PASS
AUC 0.516 +/- 0.030
p=0.367
SPP
Structure & Parameter Privacy
Dataset
18 bucket contrasts
Attack tests
structure, predicates, params
PASS
18 contrasts
AUC 0.421-0.530
Positive Control
gas-normalization ablation of the padding layer
Ablation setup
Padding OFF vs. ON
Replay attacks
same PCH/PI/SPP tests
Signal collapses
median Delta: +0.099 SPP
median Delta: +0.052 PI